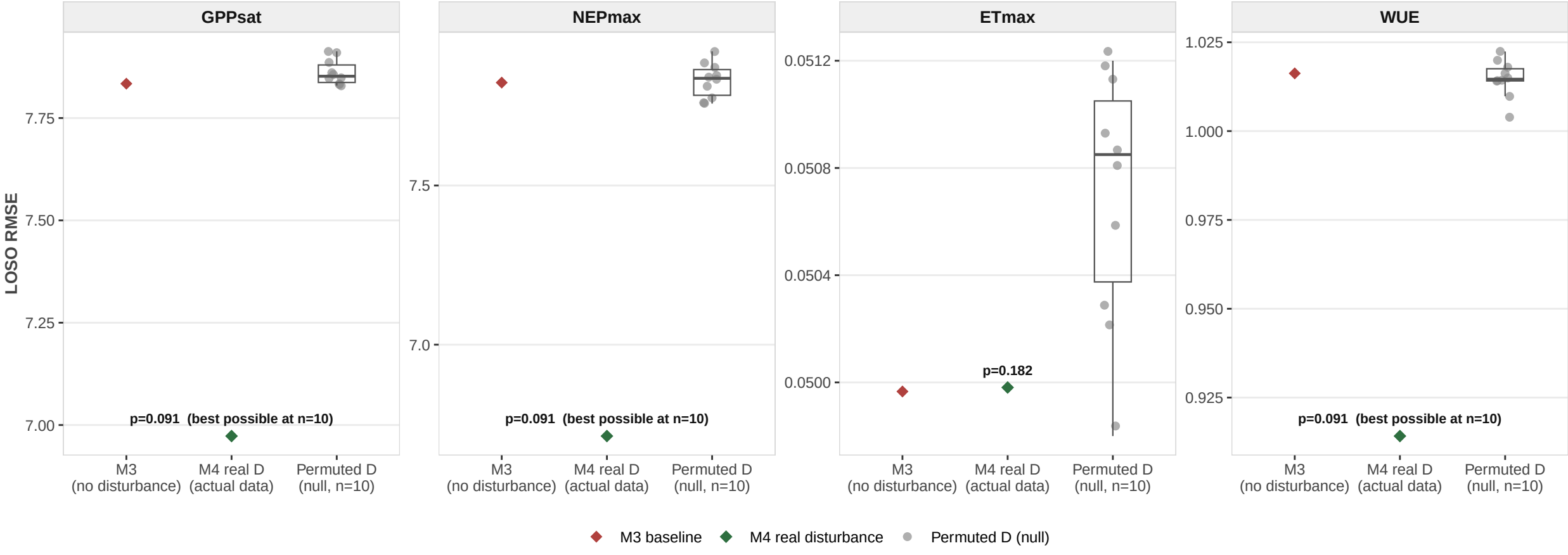
Permutation test: is the disturbance−block RMSE gain real information, or added flexibility?

XGBoost−Optuna, tc>=30 (93 sites / 395 site−years), 12−month window, LOSO CV | Null = M4 retrained with the disturbance block's rows block−permuted (10 reps)



p = (# permuted reps with RMSE <= real M4 RMSE + 1) / (n_perm + 1), one−sided. p=0.091 = real M4 beat all 10/10 permutations (the floor at n_perm=10).
ETmax's real M4 falling inside the null cloud is expected: disturbance did not improve ETmax in the main results either − an internal negative control.